Something about tensor networks, symmetries, anyons and categories.

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July 30, 2026

Contents

1 Introduction

Tensor Networks allow the efficient representation and computation of quantum states for large N using methods such as DMRG. Symmetric Tensor Networks yield a speedup over traditional methods for appropriate Hamiltonians. Furthermore, we would also like to represent *Anyonic* states and Hamiltonians by these operations, and since some anyon species are universal for quantum computation, we would like to represent circuits as well. It is folklore that tensor networks and quantum systems in general are goverened by category-theoretic constructions.

In this thesis we give a formal overview how all these concepts fit into the same categorical framework while presenting a review of the existing literature, i.e. we give an interpretation of the results of Singh and Vidal [SV12], Weichselbaum [Wei12b], Pfeifer et al. [Pfe+10], Unfried [Unf24], and Devos and Haegeman [DH25], embed them with a rigorous categorical theory and provide missing proofs. Due to the interconnectedness with topological order, we phrase them in the language of [KZ22].

1.1 Thesis Outline

In Section 2 we review the technical literature which is shared by the chapters (sections?). We give accounts on symmetries, representation theory, fermionic networks, anyons and categories. Theorem 6.14 presents the main theorem that is the backend of our tensor system. While we will state the first 2 sections with scarce proofs, chapter **??** will be quite rigorous and supercede all statements from the previous sections.

Maybe we will present a outlook on Modular Categories

2 Background

Literature 2.1 (Tensor Networks). The modern theory of *tensor networks* shaped out of the study of *renormalization* and quantum many-body systems, initially with the *density-matrix-renormalization-group* of White [Whi92; Whi93] and later extended by Ostlund and Rommer [OR95] as a variational method (ansatz?) in the form of *matrix product states* (MPS) (Maybe this is actually due to [FNW92]). This was further generalized to 2 dimensions with the introduction of *projected entangled-pair states* (PEPS) [VC04]. *Multiscale-entanglement renormalization ansatz* (MERA) [Vid08] can be considered as an extension of MPS allowing for long-range entanglement (this is probably wrong). Whats missing? Holography [SMB18] but there are earlier ones.

In the purely mathematical view as a network of tensor contractions however, both concept and the notation we use today were introduced by [Pen71] under the name of *tensor-systems*.

While we introduce this concept classically and endow it with symmetry as a start, we will start concerning ourselves with a categorical embedding of tensor networks in what is commonly reffereed to as a *fusion-category* (Lit ???).

Literature 2.2 (Symmetric Tensor Network). Originally, the idea to exploit symmetries of the hamiltonian for a speedup was proposed to be used in dmrg by [MG02] (What is [SM98]? This sounds like it as well.). More general theory for *globally invariant tensor networks*, i.e. networks that are built of tensors that are globally invariant is given by Singh, Pfeifer, and Vidal [SPV10] preceeding more detailed accounts on abelian symmetries [SPV11] and nonabelian (SU(2)) symmetries in [SV12]. A similar approach to (non)abelian symmetries is given by Weichselbaum [Wei12a; Wei20]. These rely on the general idea, that under representations of some compact group G, there are on site symmetries on each tensor, i.e. $[A, U_g^{\otimes k}] = 0$ there is a generalized *wigner-eckart-theorem* for tensors with n legs of the form

$$T \in \bigoplus_a D_a \otimes V_a$$

where D_a is a matrix and V_a is a *fusion-tree*, i.e. a clebsch-gordan tensor. Reviews and specific algorithms in the SU(2) case are given by [SO20; Sch20; Sch+20]. [Sch+20, § 8] proposes that this method is appropriate for Anyonic simulation as well (although this is done earlier by singh & pfeiffer.). Symmetries as categorified networks, as we will present in **??**, are explained in [Unf24] and [DH25] and in this work in section 5.

For a list of software implementing symmetries see **??**.

Literature 2.3 (Anyons). *Anyons* as quasiparticles with exotic exchange statistics were originally propsed by [Wil90] [LM77]. Anyons are usually described as *defects* in a (topologically orderd?) medium of dimension two (technically codimension after [**SS23AnyonsInTedK**]). The exchange statistics of N particles on this medium M is goverend by the *configuration space*

$$\mathrm{Conf}_N(M) = \{(x_1, \dots, x_N) \in M^N : x_i \neq x_j\} \setminus S_N.$$

The statistics of the particle exchange is given by the representation of $\pi_1(\mathrm{Conf}_N(M))$ (the fundamental (homology?) group). If $M = \mathbb{R}^2$, this corresponds to the *braid group* on N strands B_N. If the dimension of $M \geq 3$ this corresponds to the symmetric group S_N. If these representations are unitary and the (ground state?) is degenerate, this allows for *topological quantum computation* (c.f. Literature 2.7).

Literature 2.4 (Anyonic Spin Chains and Tensor Networks)**.** (Check if this is the first). A first appearance of *Anyonic Tensor Networks* in terms of spin chains is given by Feiguin et al. [Fei+07] as the *golden chain*, which realizes a interacting 2-local hamiltonian on a n Fib-anyon lattice. This is reviewed in [Tre+08] with generalizations to 3-local and longer range hamiltonians. A general account outside of the Fib-model can be found in [Gil+13] and [BG17] (Check these again.). We can consider these approaches as a sort of manualization of the tensor network approach. Anyonic tensor networks are first introduced by [Pfe+10] and [Pfe12], developed out of the symmetry methods from section 3 and both the *isotopy* rules and anyonic operators defined in [BSS08]. Furthermore, stuctures such as an anyonic MERA are introduced.

We also need to discuss the topic of *anyonic entanglement entropy*, as given in [Pfe14; BKP17; Pfe12].

Literature 2.5 (Simulation of Anyons and String Nets)**.** This should probably go in the previous section. [Liu+22] [Kir+23]

Literature 2.6 (Anyons as Categories)**.** The correspondence between the *physical* treatment of anyons and the *categorical treatment* of anyons as a *braided fusion category* (modular functor?) was originally descibed by Kitaev [Kit06, § E, § 8] based on previous work in quantum field theory already making use of modular categories and their diagrammatic calculus such as [RT91]. Further treatments are found in [**wang**] and [**simon-nayak-das-sarma**]. A modern treatment of the realization through the Configuration space of n-anyons if given by [Val21] and in terms of modular functors by [Bur16].

Literature 2.7 (Topological Quantum Computation). Fault tolerant *Topological Quantum Computation (TQC)* using anabelian anyons was originally proposed by Kitaev [Kit03, § 7] (Originally published in 1997). Also [Fre98] and .... ??? has shown that there are nonabelian anyon models that are universal [DN05] [Moc03] [Moc04].

[MSS24] gives nicer literature. [Sim23] is a great textbook account from one of the original authors on quantum compiling and computing. [Nay+08] is also one of the original accounts. This does not need to be any longer. Maybe mention categorical computation in general? but idk what htis is yet [KZ23].

Literature 2.8 (Quantum Compiling). Some (Many?) anyon theories are universal for quantum computation. The *Solovay-Kitaev theorem* [Kit97][DN05] yields an efficient algorithm for approximating any quantum gate using only basic gates from a universal set. Here we can consider 1.) The representation of braid circuits as a tensor nework operation and 2). A natural embedding of injection weaves as proposed by Simon et al. [Sim+06] where 2 anyons are braided in their composite fusion channel by realizing this operation as a tensor reshape.

Literature 2.9 (Categorified Quantum Mechanics)**.** There are multiple approach to a diagramatic and categorified approach to quantum mechanics. Generally one wants both spaces and maps represented by a framework that allows for parallel and entangled operations, i.e. has a tensor product structure. [Sel11] gives a review of this [VVW18] does for tensor networks specifically.

TODO: find the XZ-calculus literature by Kissinger? Specific examples such as XZ-calculus in the context of TQC is handled by [AK23].

TODO: How to mention that this results in or from $\mathrm{Hilb}_{\mathrm{fd}}$? Does it? The work from selinger et.al does not yet include anyons. For this, one needs Literature 2.6.

Also, take [SS26, § 1] for a description of categorified mechanics. This stats that structure for *entangled* processes is given by the monoidal category of complex vector spaces.

$$\begin{tikzcd}
\overset{\text{parallel quantum processes}}{\text{entanglement, no-cloning}} & (\mathrm{Mod}_{\mathbb{C}}, \otimes, \mathbb{C}) \arrow[r, "\dagger\text{-structure}"] & (\mathrm{Hilb}_{\mathrm{fd}}, \otimes, ?) \\
\overset{\text{quantum processes}}{\text{superpositions by}\oplus} & \mathrm{Mod}_{\mathbb{C}} \arrow[u] \arrow[r, "\dagger\text{-structure}"] & \mathrm{Hilb}_{\mathrm{fd}} \arrow[u]
\end{tikzcd}$$

The categorical language specifically yield interpretations of theorems such as no-cloning and no deleting [Abr12].

Additionally, many of the concepts of topological order are governed by category theory [KZ22], as well as the topic of higher symmetries (should this be its own Lit sect if relevant? It is for MPO symmetries [DT24] [Kon+20]).

2.1 Tensors (Everything here is $fdVect_k$)

Generally, we want to regard a tensor as a thing equipped with legs, where each legs holds a vector space. But we also want to be able to regard tensors as operators such as Hamiltonians.

Lets list some definitions of Tensor. A Tensor T can be regarded as an element of a tensor product space

$$T \in \bigotimes_{j \in J} V_j$$

where each V_j is a finite-dimensional vector space and we say T is a $|J|$-legged tensor.

Given !finite! dimensional Vector spaces V and W, we have the identity that $V \otimes W^* \equiv Hom(W, V)$. This is called the *hom-tensor duality.* (TODO: is this canonical? What about the other way around?)

In a basis expansion, this would relate the expansion of $V \otimes W^*$ as

$$T = \sum_i v_i \otimes f_i$$

where $f_i \in W^* : W \to k$ with the map $T \in \mathrm{Hom}(W, V)$ as

$$T = (w \mapsto \sum_i f_i(w) v_i).$$

Now, if we have a Tensor $T \in V_1 \otimes \cdots \otimes V_n$, and since for any Vector space V, $(V^*)^* = V$, we can peel off any number of V_is, take their duals and represent T as a map

$$T : V_n^* \otimes \cdots \otimes V_{k+1}^* \to V_1 \otimes \cdots \otimes V_k$$

Note that we present the domain here in a contravariant fashion due to the contravariance of the dual.

Lets look at an example, let $t \in V_1 \otimes V_2 \otimes W_2^* \otimes W_1^*$. We can represent as an operator in dirac notation:

$$t = \sum_{i_1, i_2, j_1, j_2} T^{i_1 i_2}_{j_1 j_2} \left| v^1_{i_1} \right\rangle \left| v^2_{i_2} \right\rangle \left\langle w^1_{j_1} \right| \left\langle w^2_{j_2} \right|$$

where $|v_i\rangle$

Graphically, we represent a general tensor

$$T : (W_n^* \otimes \cdots \otimes W_1) \to (V_1 \otimes V_2 \otimes \cdots \otimes V_n)$$

as follows, where we choose incoming arrows for duals in the domain and outgoings for normal spaces, and on the bottom we choose incoming for duals and outgoing for normals.

3 Symmetric Tensor Networks

This section develops the theory underlying *symmetric tensor networks*, specifically for nonabelian groups. We give an informal overview on the necessary representation theoretic concepts in Section 3.1. We first give some motivation with the worked example of the XXZ-Hamiltonian and then show how the same is achieved in a representation theoretic setting. Here we are mostly interested in the *speedup* obtained by exploiting the emergend block-structure of symmetric operators. Most of the proofs and formal definitions required for this formalism are delegated to Section 6, however there will still be repetitions of some of the theory.

3.1 Representation theory

In the following and the entire section, let G be a compact group and V, W be finite dimensional vector spaces. We mainly follow the textbook by **woit2016**.

Definition 3.1 (Representation)**.** A *representation* of a compact group G over a vector space V is a group homomorphism

$$\rho : G \to \mathrm{GL}(V), \quad \rho(g)\rho(h) = \rho(g \cdot h), \quad \rho(e) = id_V$$

The *representation space* is denoted by the tuple (ρ, V) but usually we shorten this to just V.

Equivalently we can state the definition of a representation of G as imposed by its action on V, given as a map

$$\alpha : G \times V \to V, \quad (g, v) \mapsto g \cdot v \tag{1}$$

such that this is linear, todo. I think we get the equality to the hom def if we set $g \cdot v = \rho(g)v$

Definition 3.2 (Intertwiner)**.** A linear map $A : V \to W$ is an *intertwiner*, if under representations $\rho_V : G \to \mathrm{GL}(V)$ and $\rho_W : G \to \mathrm{GL}(W)$ it holds that

$$A \circ \rho_V(g) = \rho_W(g) \circ A \quad \forall g \in G.$$

Denote the space of all intertwiners as $\mathrm{Hom}_G(V, W)$.

Definition 3.3 (Unitary representation)**.** If our vector space V is equipped with a (Hermitian?) inner product

Definition 3.4 (Irreducible representation (Woit 16))**.** A representation ρ is an *irrep* if there are no $W \subset V$ such that $(\rho_{|W}, W)$ is a representation.

Definition 3.5 (Tensor product representation)**.** Given reps (ρ_V, V) and (ρ_W, W), the *tensor product representation* $(\rho_{V\otimes W}, V \otimes W)$ is defined by its action on the elements

$$\rho_{V\otimes W}(g)(v \otimes w) = \rho_V(g)v \otimes \rho_W(g)w$$

Definition 3.6 (Direct sum representation)**.** As above, the *direct sum representation* $(\rho_{V\oplus W}, V \oplus W)$

$$\rho_{V\oplus W}(g) = \begin{pmatrix} \rho_V(g) & \\ & \rho_W(g) \end{pmatrix}$$

Definition 3.7 (Isotypic / Semisimple decomposition)**.** Each semisimple representation V decomposes as

$$V \cong \bigoplus_{i\in I} V_i^{\oplus n_i} \cong \bigoplus_{i\in I} \mathbb{C}^{n_i} \otimes V_i \cong \bigoplus_{i\in I} V_i \otimes \mathbb{C}^{n_i}$$

by grouping isomorphic simples. Is n_i unique?

Due to a theorem by Peter & Weyl [<**empty citation**>], the unitary representations of any compact G are semisimple.

Therefore, it must hold for any irreps R_a, R_b of G that their tensor product is completely reducible with the decomposition

$$R_a \otimes R_b \cong \bigoplus_{c\in I} R_c^{\oplus N_{ab}^c},$$

with N_{ab}^c called *fusion-multiplicities*. If

$$N_{ab}^c \leq 1 \quad \forall a, b, c$$

the decomposition is said to be *multiplicity-free*.

Note as well that the appearance of the *direct sum* here (as discussed in eq. (14)) is fully characterized (carefull how i phrase this) by its injection and projection maps, for which we provide a graphical calculus.

$$\iota_{c,\mu}^{ab} : R_c \longrightarrow R_a \otimes R_b \quad \text{and} \quad \pi_{ab}^{c,\mu} : R_a \otimes R_b \longrightarrow R_c \quad \text{for } \mu \in 1, \ldots, N_{ab}^c$$

(2)

These maps are more commonly known as *clebsch-gordan coefficients*.

On the operators between tensor products, i.e. $T \in \mathrm{End}(R_a \otimes R_b)$ this yields a formula

$$T \cong \bigoplus_c \mathrm{id}_{R_c} \otimes \mathbb{C}^{N_{ab}^c}$$

Example 3.8. The simple objects of $G = \mathrm{U}(1)$ are given by the one-dimensional representations

$$\chi_n : U(1) \longrightarrow \mathbb{C}^\times, \quad e^{i\theta} \mapsto e^{in\theta}$$

and each representation of $\mathrm{U}(1)$ decomposes as

$$V \cong \bigoplus_{n \in \mathbb{Z}} \chi_n^{\oplus m_n}.$$

It's fusion rule is given by

$$\chi_n \otimes \chi_m = \chi_{n+m}$$

Example 3.9. Let $G = \mathrm{SU}(2)$ and let V_i and V_j be two spin irreps of charge i and j. These combine to

$$(\pi_{V_i \otimes V_j}, V_i \otimes V_j) = \bigoplus_{k \in |i-j|}^{i+j} (\pi_{V_k}, V_k) \tag{3}$$

in what is referred to as the *Clebsch-Gordan-series*.

Example 3.10 (SU(2)-CG tensors)**.** Lets look at an example of spin irreps in the $\mathrm{SU}(2)$ theory. The combined basis of two spin-$\frac{1}{2}$ spaces is given by the collection $\{|\uparrow\rangle, |\downarrow\rangle\}^{\otimes 2}$. By the fusion rule eq. (3), this decomposes as

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

with its accosiated basis $\{|1,1\rangle, |1,0\rangle, |1,-1\rangle, |0,0\rangle\}$. The change of base tensors acting between these bases are precisely the Clebsch-Gordan tensors as defined in ???

3.2 Examples in symmetric hamiltonians.

The motivation for constructing symmetric tensor networks is as follows: Assume we have a (unitary) representation of a group G denoted $U(g)$. Furthermore assume we have a hamiltonian that commutes with $[U(g)^{\otimes L}, H] = 0$. Then there are different presentations of H with respect to a certain basis by *simulateneous diagonalization*[HJ12, Thm 1.3.12], as shown in Figure 1. Notably these presentation induce a *block-sparse* structure on the respective operators.

We now show how to obtain this block-sparse structure and more by example of the XXZ hamiltonian:

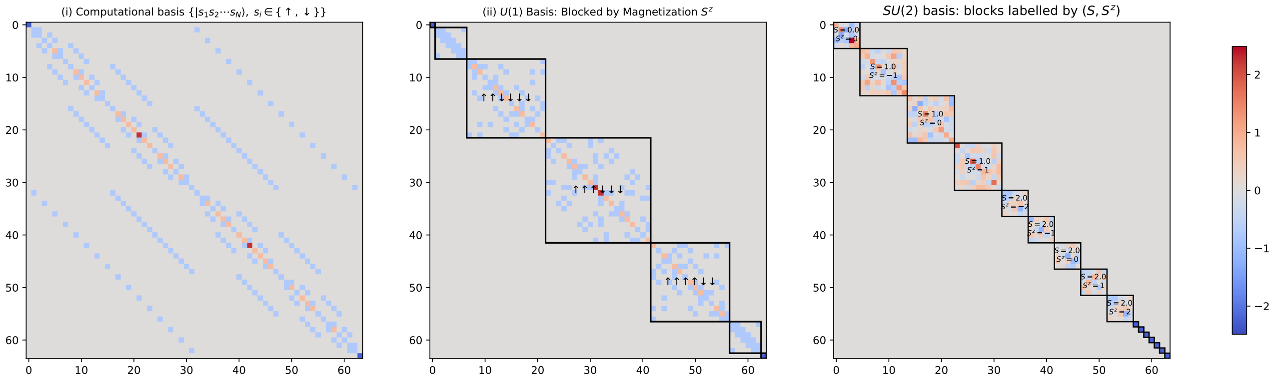


Figure 1: Three different presentations of the XXZ Hamiltonian for $N = 6$. (i): In the default computational basis of spin vectors. (ii) In a reordering of the computational basis into equal spin up (down) classes, i.e. no spin up, one spin up etc..., here U is simply a permutation matrix. (iii) After changing the basis into the (S^2, S_Z) basis.

Definition 3.11. The *XXZ Hamiltonian* on N sites is

$$H = J \cdot \sum_{i=1}^{N-1} (\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y + \Delta \sigma_i^z \sigma_i^z), \tag{4}$$

acting on the basis $|s_1 s_2 \ldots s_n\rangle$ where $s_i \in \{\uparrow, \downarrow\}$.

This operator admits both a U(1) symmetry, and a SU(2) symmetry for $\Delta = 1$. We elaborate on the (simpler) abelian case first:

The representations of U(1) here are induced by the operator

$$S^z_{\text{tot}} = \frac{1}{2} \sum_{i=1}^{N-1} \sigma_i^z \quad \text{with action} \quad U(\theta) : \theta \in \mathrm{U}(1) \mapsto e^{-i\theta S^z_{\text{tot}}}$$

It can be shown that H is invariant under $U(\theta)$:

$$U(\theta) H U(\theta)^\dagger \underset{\text{suffices by A.1}}{\iff} [H, S^z_{\text{tot}}] = 0. \tag{5}$$

Diagonalizing both operators yields a block structure as in Figure 1(ii). Note that change of base matrix in this case is given by a permutation matrix.

If we want to state this in terms of irreps, we have to realize that $\mathbb{C}^2 \cong \chi_{-1} \oplus \chi_1$. Then we can state the full space as

$$(\mathbb{C}^2)^{\otimes 6} \cong \chi_{-3} \oplus \chi_{-2}^6 \oplus \chi_{-1}^{\oplus 15} \oplus \chi_0^{\oplus 20} \oplus \chi_1^{\oplus 15} \oplus \chi_2^{\oplus 6} \oplus \chi_3$$

and so we should expect 7 blocks in total, just as given in Figure 1.

In the SU(2) case, we have the generators

$$S^\alpha_{tot} = \frac{1}{2} \sum_{i=1}^{N} \sigma_i^\alpha, \quad \alpha \in \{x, y, z\}$$

The representation here is $U(R) = e^{i\theta n \cdot S_{tot}}$ and the Hamiltonian is invariant under

$$U(R) H U(R)^\dagger = H, \quad R \in \mathrm{SU}(2)$$

Since

$$[S^x_{tot}, S^y_{tot}] = i S^z_{tot}$$

we need a new operator for the simulatenous diagonalization

$$S_{tot} = \sum_{a \in \{x,y,z\}} (S^a_{tot})^2$$

and we need to verify this is indeed the correct operator TODO.

The eigenvalues of S^2_{tot} are $S(S+1)$ where S are the half integers.

Again, we can solve this by diagonalizing all 3 operators? Do this calculation as in the .ipynb and put it in the appendix.

Or we can do the representation theoritical approach. Let V_j denote the spin-j irrep of $\mathrm{SU}(2)$. We have that $\mathbb{C}^2 \cong V_{\frac{1}{2}}$ and by **??** that

$$(\mathbb{C}^2)^{\otimes 6} \cong V_{\frac{1}{2}}^{\otimes 6} \cong 5V_0 \oplus 9V_1 \oplus 5V_2 \oplus V_3 \cong \mathbb{C}^5 \otimes V_{\frac{1}{2}} \oplus \mathbb{C}^9 \otimes V_0 \oplus \mathbb{C}^5 \otimes V_2 \oplus V_3$$

Now our operator is a morphism in $End(V_{\frac{1}{2}}^{\otimes 6})$ and thus by Schurs lemma[1]

$$H \in \bigoplus_{S\in 0,1,2,3} Mat_{m_S\times m_S} \otimes I_{2S+1}(\mathbb{C}) \cong \bigoplus_{S\in 0,1,2,3} I_{2S+1} \otimes Mat_{m_S\times m_S}(\mathbb{C}) \tag{6}$$

So we expect 20 blocks? 5 have dimension 1, 9 have dimension 3, 5 have dimension 5 and 1 has 7.

To get an interpretation of the matrix coefficients in (3), consider a tiny Hamiltonian on $N = 2$ spins

$$H = \frac{1}{2}(\sigma_1^x\sigma_2^x + \sigma_1^y\sigma_2^y + \sigma_1^z\sigma_2^z) = S_1 \cdot S_2$$

This acts on the space $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ and thus on the basis $\{|\uparrow\rangle, |\downarrow\rangle\}^{\otimes 2}$ This has two distinct fusion outcomes, V_0 and V_1. The basis on the rhs is given by $|1,1\rangle, |1,0\rangle, |1,-1\rangle . |0,0\rangle$, and we want to reformulate $S_1 \cdot S_2$ in the total spin basis. Lets unfold this definition by

$$S^2 = S_1^2 + S_2^2 + 2(S_1 \cdot S_2) \implies S_1 \cdot S_2 = \frac{1}{2}(S^2 - S_1^2 - S_2^2).$$

Since $S_i^2 = s(s+1)$ is fixed for $s = \frac{1}{2}$ here, we ought to write

$$H = \frac{J}{2}(S^2 - 3/2)$$

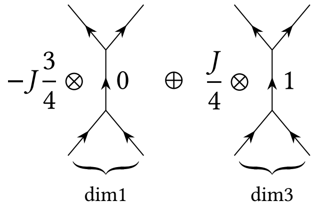
and this again splits into the sectors for $S = 0$ and $S = 1$

$$c_0 = \frac{J}{2}(0 - 3/2) = -\frac{3}{4}J \quad \text{and} \quad c_1 = \frac{J}{2}(2 - 3/2) = \frac{1}{4}J$$

which are exactly the two degrees of freedom we get by the direct sum. So we expect our operator to look like

$$\begin{pmatrix} c_0 & 0 \\ 0 & \mathrm{id}_3 \otimes c_1 \end{pmatrix}.$$

By direct isomorphism, we can write this as



(7)

[1] These are related by a permutation matrix. Proof? Is this the X from [DH25]? We prefer the second presentation as this realizes more blocks.

3.3 Symmetric Tensors

Let G be a compact group. Consider a tensor T living in the tensor product space of its legs:

$$T \in V_1 \otimes \cdots \otimes V_n \otimes W_m^* \otimes \cdots \otimes W_1^*$$

Let $\rho_i : G \longrightarrow \mathrm{GL}(V_i)$ be the linear representation of G on each vector space V_i, and let $\sigma_j^* : G \longrightarrow \mathrm{GL}(W_j^*)$ be the dual representation on W_j^*.

The total action of the group G on the entire tensor product space is given by

$$\Pi : G \longrightarrow \mathrm{GL}\left(\bigotimes_i V_i \otimes \bigotimes_j W_j^*\right)$$
$$\Pi(g) = \rho_1(g) \otimes \cdots \otimes \rho_n(g) \otimes \sigma_m^*(g) \otimes \cdots \otimes \sigma_1^*(g)$$

A tensor T is defined to be a **symmetric tensor** (or G-invariant tensor) if it is invariant under simultaneous action on all legs for all $g \in G$ [SPV10]:

$$\Pi(g)T = T \quad \forall g \in G$$

Recall that by the isomorphism $\mathrm{Hom}(V, W) \cong W \otimes V^*$ as

$$w \otimes f \mapsto (v \mapsto f(v)w).$$

We can push the product representation through this isomorphism as

$$\pi(g)(w \otimes f) = \rho_W(g)w \otimes \rho_{V^*}(g)f = \rho_W(g)w \otimes (\rho_V(g)^{-1})^T f = \rho_W(g)w \otimes f(\rho_V(g)^{-1})$$
$$\cong v \mapsto f(\rho_V(g)^{-1}v)\rho_W(g)w$$

By interpreting T as a map

$$T : W_1 \otimes \cdots \otimes W_m \longrightarrow V_1 \otimes \cdots \otimes V_n$$

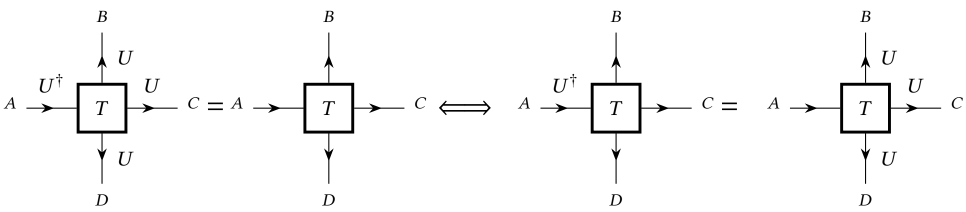
it follows directly from the definition of global invariance that the tensor must commute with the group action and we get the following definition of a symmetric tensor:

Definition 3.12 (symmetric tensor)**.** A *symmetric tensor* is an intertwiner of the representations of the group G:

$$\bigotimes_{j=1}^{m} \sigma_j(g) \circ T = T \circ \bigotimes_{i=1}^{n} \rho_i(g) \quad \forall g \in G,$$

which we denote by

$$T \in Hom_G(W, V).$$



Globally invariant
$T \in \mathrm{Hom}(A, B \otimes C \otimes D)$

is an intertwiner
$T \in \mathrm{Hom}_G(A, B \otimes C \otimes D)$

Corollary 3.13 (Networks of symmetric tensors are symmetric)**.** *Since a tensor network is built by composition and application of the* (co)*ev maps, both of which are intertwiners, the full network is also an intertwiner. Important there is that this entire discussion is only about FdVect or FdHilb for now, but i think the same applies to any braided category.*

Definition 3.14. A *Matrix Product State* is a quantum state $|\psi\rangle$ with n legs written as the contraction of n trivalent tensors $\{A^{[i]}\}_{i\in[n]}$. Formally this reads

$$|\psi[A]\rangle = \sum_{i_1,\dots,i_n} \mathrm{Tr}\Big(A^{[1]}_{i_1} A^{[2]}_{i_2} \cdots A^{[n]}_{i_n}\Big) |i_1 i_2 \cdots i_n\rangle \,,$$

where $A^{[i]}_{i_k} \in \mathbb{C}^{D_i \times D_{i+1}}$. The elements of the collection $\{D_i\}_{i\in[n]}$ is referred to as the *bond-dimensions*.

Note that this is a state with closed boundaries. If we want open boundaries we have to replace the trace with appropriate end vectors (nice source?).

Example 3.15. We can represent each product state $|\Psi\rangle = |\psi\rangle^{\otimes n}$ as a MPS with bond dimension 1.

Theorem 3.16 (Fundamental Theorem of MPS)**.** *If two MPS states generated by A and B respectively. Figure out what we can do with this.*

Proof [Cir+17],[VVW18], [Per+08].

3.4 Algorithms for block-sparse tensors

4 Fermionic Tensor Networks

Here we follow [Bul+17; Gao+25; Mor+25] primarily. These papers present the theory of fermionic networks over the category of *super vector spaces*, denoted sVect, which is also equal to a $\mathbb{Z}_2$-graded Vector space $\mathrm{Vec}_{\mathbb{Z}_2}$ as we will see in section **??**. Each $V \in$ sVect admits a decomposition into *odd* and *even* parity sectors

$$V = V_0 \oplus V_1.$$

Denote the tensor product in sVect by $\otimes_{\mathbb{Z}_2}$. Since fermions posess a non-trivial *exchange statistic*, i.e. exchanging two fermions incurs a phase factor of -1 on the wave function, the graded tensor product encodes this under its canonical braiding [2] also called the *Koszul sign rule*

$$|i\rangle \otimes_{\mathbb{Z}_2} |j\rangle = (-1)^{|i||j|} |j\rangle \otimes_{\mathbb{Z}_2} |i\rangle \,. \tag{8}$$

5 Anyonic Tensor Networks

We need to introduce anyons here, or somehow build it into the review of the golden chain.

This chapter investigates methods that define tensor networks on physical systems that constitute *anyons*. We introduce the fundamental theory of anyons algebraically (Section 5.1), and make use of the algebraic rules as the building blocks for our theory. Section 5.2 gives examples of how the statistical properties of anyons change with respect to it's ground state. Then, in **??**, we provide a worked examples of how anyons are treated on *spin-chains*, which motivates some aspect of the tensor networks approach. TODO tensor network chapter. One of the primary motivations for the study of anyons are their universality for quantum computation. In **??** we provide an interpretation of how both normal braiding circuits and a variation in terms of *weaves* can be simulated by the tensor network theory.

5.1 Algebraic Theory of Anyons

Anyons are proposed as a *defect* in a 2 dimensional space (manifold? compact?) with exotic exchange statistics. These exchange statistics differ from fermions in that we allow particles to pick up a arbitrary phase $e^{i\theta}$ on exchange in the abelian case, and a phase in a matrix element in the nonabelian case.

5.2 Anyons on different topologies

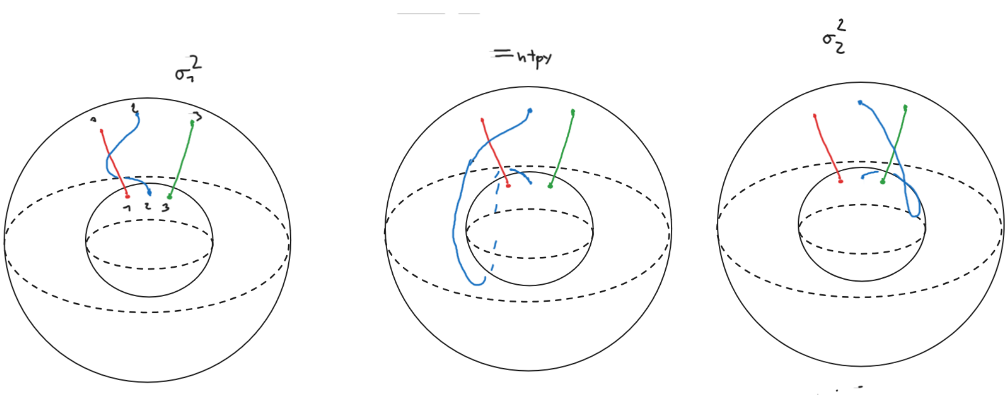


Figure 2: 3 Anyons on the sphere S^2 with the equivalence of the braids σ_1^2 and σ_2^2. From an exercisegithub in [Sim23].

Analysing this case further yields that

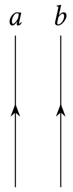
$$\pi_1(\mathrm{Conf}_3(S^2)) \cong Br(S^2) \cong \mathbb{Z}_4 \boxtimes \mathbb{Z}_3$$

[2] Check on the Nlab, the proof that this is canonical is quite copmlex?

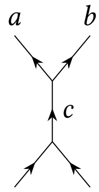
and thus finite. In fact, we know about the braid group on spheres that the order of $Br_n(S^2)$ is given by

$$\left|Br_n(S^2)\right| = \begin{cases} |\mathbb{Z}_2| = 2 & \text{if } n = 2, \text{ [FB62, p. 255]} \\ |???| = 12 & \text{if } n = 3, \text{ [nLa26, Prop 2.2]} \\ \infty & \text{if } n \geq 4. \end{cases} \tag{9}$$

Also we should do or have done isotopy before. Technically the discussion here also requires the completeness relations.

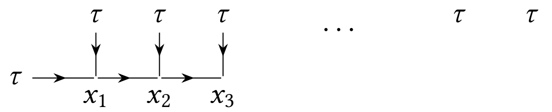


$$= \sum_{c,\mu} \sqrt{\frac{d_c}{d_a d_b}} \tag{10}$$



5.3 1D Lattice Models & The Golden Chain

In [Fei+07], anyonic chains, specifically in the Fib model, are introduced as a 1D lattice of n τ anyons, defined as the fusion space thereof.



Here, the state is fully characterized by its intermediate indeces $x_1 \dots x_{n-1}$, so we can choose a basis $|x_1 x_2 \cdots x_{n-1}\rangle$.

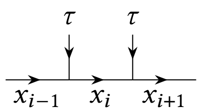
Similar to the Heisenberg-chain[3], the golden chain introduces a Hamiltonian that assigns -1 to the interaction $\tau \otimes \tau = 1$ and 0 to the interaction $\tau \otimes \tau = \tau$.

We can decompose the whole space $\mathrm{Hom}(\tau^{\otimes n+1}, \tau)$ into the local states

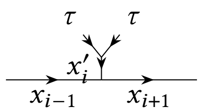
$$\mathrm{Hom}(\tau^{\otimes n+1}, \tau) = \bigoplus_{\substack{x_1, x_2, \dots, x_{n-1} \\ \text{all admissible paths}}} \mathrm{Hom}(\tau \otimes \tau, x_1) \otimes \mathrm{Hom}(x_1 \otimes \tau, x_2) \otimes \cdots \otimes Hom(x_{n-1} \otimes \tau, \tau) \tag{11}$$

with the admissability condition that there are not adjacents 1 in the sequence of x_is.

This is evaluated as a 3 local hamiltonian, as we can decompose the spin chain into local parts where two τ anyons are adjacent (check how they call it in the paper.).



$$= \sum_{x_i} [F^{x_{i-1}\tau\tau}_{x_{i+1}}]^{x_i'}_{x_i}$$



$\in \mathrm{Hom}(x_{i-1} \otimes \tau, x_i) \otimes \mathrm{Hom}(x_i \otimes \tau, x_{i+1})$ $\qquad$ $\in \mathrm{Hom}(x_{i-1} \otimes x_i', x_{i+1}) \otimes \mathrm{Hom}(\tau \otimes \tau, x_i')$

Now we can apply the interaction based on the intermediate state[4] x_i' and apply another F-move back to the chain basis.

$$H_{[i,i+1]} : |x_{i-1} x_i x_{i+1}\rangle \to \left|x_{i-1} x_i'' x_{i+1}\right\rangle \coloneqq - \sum_{x_i' \in \{1, \tau\}} [F^{x_{i-1}\tau\tau}_{x_{i+1}}]^{x_i'}_{x_i} E_{x_i'} [F^{x_{i-1}\tau\tau}_{x_{i+1}}]^{x_{i''}}_{x_{i'}}$$

Write this three local hamiltonian in an operator form

$$H_i = -J \begin{pmatrix} 1 & & & & \\ & 0 & & & \\ & & 0 & & \\ & & & \varphi^{-2} & \varphi^{-\frac{3}{2}} \\ & & & \varphi^{-\frac{3}{2}} & \varphi^{-1} \end{pmatrix}$$

[3] Note that $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$ in the Heisenberg SU(2) model.

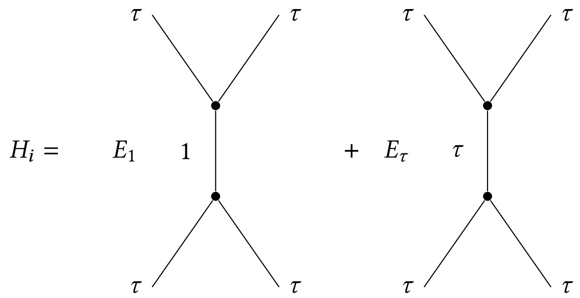
[4] Technically we need a projector here read up again.

This method is reviewed and extended in [Tre+08], with longer ranger interaction. It will become apparent that the construction given in these papers is a manualization of the tensor network constructions we will build here. Generalizations to arbitrary models are given in [Gil+13] as well as [BG17]. Generalized spin chains and dualities using module categories are given in [Eck25].

The tensor network formalism is handled in [Pfe+10], [Pfe12] and [Aye17], and in terms of categories this is done in [Unf24] and [DH25].

For the tensor network operations, we should consider the changing of the basis as an implicit operation that is performed when contracting, just like it would be in a normal einsum.

In a tensor network formalism can build this as an anyonic operator in $Hom(\tau \otimes \tau, \tau \otimes \tau)$ as



(Note this is allowed due to $\mathbb{C}$-linearity of Fib.)

In this case we can view the golden chain approach chosen above as a manualization of the tensor network process.

We can also write H_i in its operator form matrix as

$$H_i \in \bigoplus_{a \in 1, \tau} Hom(\mathbb{C}, \mathbb{C})$$

as

$$H_i = -J \quad \begin{array}{cc} & \begin{array}{cc} 1 & \tau \end{array} \\ \begin{array}{c} 1 \\ \tau \end{array} & \begin{pmatrix} E_1 & 0 \\ 0 & E_\tau \end{pmatrix} \end{array}$$

5.4 Anyonic States

For simple lattice states, [Sin+14] defines a bipartition of the lattice into anyons A and B. Then we consider the whole space under charge a_{tot} as

$$V_{a_{tot}} = \bigoplus_{N_{ab}^{a_{tot}}=1} (V_a^A \otimes V_b^B)$$

What i dont get here is why we are allowed to tensor both these sectors together without the space $V_{ab}^{a_{tot}}$ explicitely.

TODO delete all of this and introduce this over the Hom category.

Anyhow, the fundamental building blocks we are using are the vector spaces V_c^{ab} (where we refer to this as the fusion space from types a and b to c). This allows us to build the vector space of two anyonic types a, b as the sum over the fusion outcome:

$$V^{ab} = \bigoplus_{c \in Irr(C)} V_c^{ab}$$

We can directly introduce the R move (a braiding of the initial anyons) to obtain the isomorphic space V^{ba} as

$$R: \bigoplus_{c \in Irr(C)} V_c^{ab} \cong \bigoplus_{c \in Irr(C)} V_c^{ba}$$

We can also write the space $V_c^{ab} = Hom(a \otimes b, c)$ in the categorical language. Here we also need to realize that there is a difference between the isomorphism β and the action of R, which is really on the hom space

Hom$(-, c)$. So this would be a yoneda perspective? Since we know R for all $a, b, c \in C$, we also know the action of R as an isomorphism. Well technically i believe that this allows us to determine for all simples c, and then semisimplicity extends this to all subjects (this would be nice to write down) and then the yoneda embedding is enough to know the whole space. Similarly, in our Rep(G) categories (which i actually belive to be symmetric? Find this in etingof)

The algebra-category correspondence in the theory of anyons was first noted by [Kit06, App E]

5.5 Topological Quantum Computation

Computing with nonabelian anyons requires a universal gate set built from representations of the *braid-group*:

$$\sigma_i \sigma_j = \sigma_j \sigma_i, \quad |i - j| \geq 2 \tag{12}$$

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}, \quad i \in [1, n-2] \tag{13}$$

and also the *pure-braid-group*:

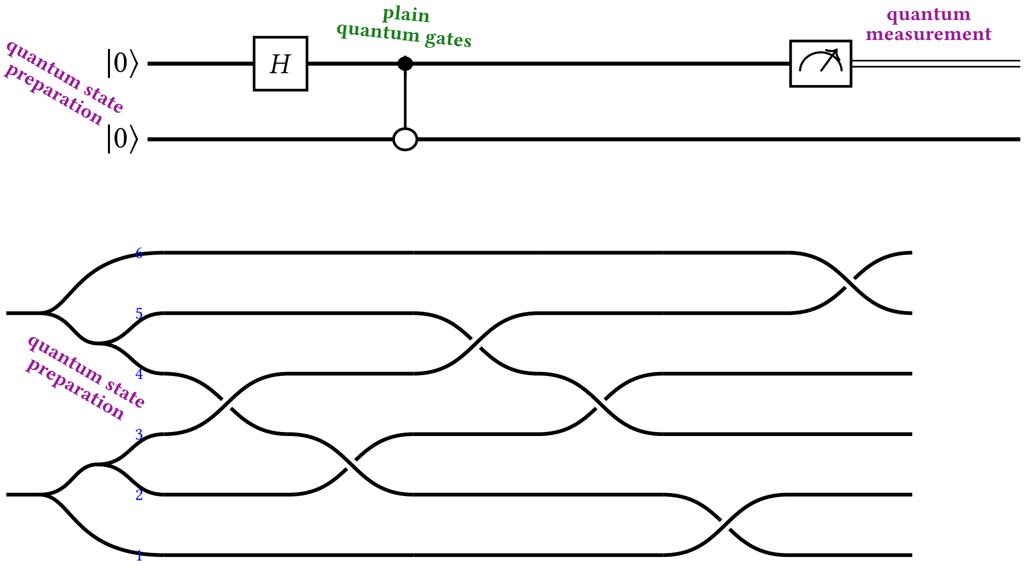


Figure 3: A standard 2-qubit quantum circuit and its lowering to a braid-circuit with inection weaves (missing yet).

6 Categorical Networks

The goal of this chapter is to provide a viewpoint on the previous chapters from the perspective of categories. In Section 6.1 we provide a short introduction to category theory, from the perspective of quantum parallelism, and the structures we have already met in the discussion on representations and anyons. In Section 6.2 we give a short review of how familiar structures such as the no-cloning theorem arise from the categorification of quantum mechanics. In Section 6.3 we provide the core theory in terms of the decomposition, which is crucial to previous chapters. Naturally, this incurs a rigorous explanation of fusion trees.

6.1 Introduction to Category Theory

The general idea of a category is that it admits a sort of *interface*. As noticed, there are many similarities in data and structure between the entities we already dealt with, i.e. the tensor product of vector space, the tensor product of representations, of Hilbert spaces etc.... We can view all these concepts as a category with additional constraints added.

Definition 6.1. A *category* C consists of

- a set of *objects* $\mathrm{ob}(C)$,
- for every pair $A, B \in \mathrm{ob}(C)$, a *hom-set* $\mathrm{Hom}_C(A, B)$ of *morphisms* $f : A \longrightarrow B$,
- for every $A \in \mathrm{ob}(C)$, an *identity morphism* $\mathrm{id}_A : A \longrightarrow A$,
- for every triple $A, B, C \in \mathrm{ob}(C)$, a *composition* map

$$\circ \;:\; C(B,C) \times C(A,B) \longrightarrow C(A,C),$$

such that composition is *associative*, i.e. $(f \circ g) \circ h = f \circ (g \circ h)$ whenever defined, and *unital*, i.e. $f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f$ for all $f : A \longrightarrow B$.

Note that some categories such as $\mathrm{Rep}(G)$ and $\mathrm{Vec}_{\mathbb{C}}$ admit more structures than ordinary categories. For fdVect, the hom spaces of linear maps $V \longrightarrow W$ have a natural identification as a vector space as well. The composition map is $\mathbb{C}$-bilinear. This immedeately motivates the following definition:

Definition 6.2. A *$\mathbb{C}$-linear category* endows each of its hom-sets with a module structure, such that the composition is bilinear. That is

$$\circ \;:\; \mathrm{Hom}_C(B,C) \times \mathrm{Hom}_C(A,B) \longrightarrow \mathrm{Hom}_C(A,C)$$

is $\mathbb{C}$-bilinear: for all $f, f' : B \longrightarrow C, \quad g, g' : A \longrightarrow B$, and $c \in \mathbb{C}$,

$$(f + f') \circ g = f \circ g + f' \circ g, \qquad f \circ (g + g') = f \circ g + f \circ g', \qquad (cf) \circ g = c(f \circ g) = f \circ (cg).$$

We will now explore how to use category theory to model quantum processes. We want this to be able to do a list of things. First, given two systems A and$B$, we want the combined system to be an object as well. Given processes, i.e. operators on each individual system $f : A \longrightarrow A'$,$g : B \longrightarrow B'$, there should be a combined process. Additionally, structures such as entanglement and no-cloning should be readily apparent in this theory.

Consider the tensor product $\otimes_{\mathbb{C}}$ of vector spaces V,W. We can define this using a *universal property* ... TODO.

This should motivate the following definition:

Definition 6.3. A *monoidal category* C is a category that admits a unit object $I \in ob(C)$ and a bifunctor $\otimes : C \times C \longrightarrow C$, as well as 3 constraining isomorphisms

$$\rho_A : A \otimes I \cong A \qquad \lambda_B : I \otimes B \cong B \qquad \alpha_{ABC} : (A \otimes B) \otimes C \cong A \otimes (B \otimes C)$$

such that the following diagrams commute:

$$\begin{tikzcd}
(A \otimes I) \otimes B \arrow[rr, "\alpha_{A,I,B}"] \arrow[dr, "\rho_A \otimes \mathrm{id}_B"'] & & A \otimes (I \otimes B) \arrow[dl, "\mathrm{id}_A \otimes \lambda_B"] \\
& A \otimes B &
\end{tikzcd}
\qquad
\begin{tikzcd}
((A \otimes B) \otimes C) \otimes D \arrow[rr, "\alpha_{A,B,C} \otimes id_D"] \arrow[d, "\alpha_{A \otimes B,C,D}"'] & & (A \otimes (B \otimes C)) \otimes D \arrow[d, "\alpha_{A,B \otimes C,D}"] \\
(A \otimes B) \otimes (C \otimes D) \arrow[dr, "\alpha_{A,B,C \otimes D}"'] & & A \otimes ((B \otimes C) \otimes D) \arrow[dl, "id_A \otimes \alpha_{B,C,D}"] \\
& A \otimes (B \otimes (C \otimes D)) &
\end{tikzcd}$$

For simplicity we denote a monoidal category as the sextuple $(C, \otimes, I, \alpha, \lambda, \rho)$.

Definition 6.4. A *braided monoidal category* is a monoidal category $(C, \otimes, I, \alpha, \lambda, \rho)$ equipped with an additional structural natural isomorphism, the *braiding*

$$\beta_{A,B} : A \otimes B \;\cong\; B \otimes A,$$

natural in A and B, such that the following two *hexagon* diagrams commute for all $A, B, C \in C$:

$$\begin{tikzcd}
(A \otimes B) \otimes C \arrow[r, "\alpha"] \arrow[d, "\beta_{A,B} \otimes \mathrm{id}_C"'] & A \otimes (B \otimes C) \arrow[r, "\beta_{A,B \otimes C}"] & (B \otimes C) \otimes A \arrow[d, "\alpha"] \\
(B \otimes A) \otimes C \arrow[r, "\alpha"] & B \otimes (A \otimes C) \arrow[r, "\mathrm{id}_B \otimes \beta_{A,C}"] & B \otimes (C \otimes A)
\end{tikzcd}
\qquad
\begin{tikzcd}
A \otimes (B \otimes C) \arrow[r, "\alpha^{-1}"] \arrow[d, "\beta"'] & (A \otimes B) \otimes C \arrow[r, "\beta"] & C \otimes (A \otimes B) \arrow[d, "\alpha^{-1}"] \\
A \otimes (C \otimes B) \arrow[r, "\alpha^{-1}"'] & (A \otimes C) \otimes B \arrow[r, "\beta"'] & (C \otimes A) \otimes B
\end{tikzcd}$$

Definition 6.5. A braided monoidal category is *symmetric* if $\beta_{B,A} \circ \beta_{A,B} = \mathrm{id}_{A \otimes B}$ for all A, B.

Definition 6.6. Given a category C with objects $\{a_n\}$, an object $\bigoplus_{i=1}^n a_i \in C$ is called the *direct sum*. This is equipped with the injection and projection map

$$\iota_i : a_i \to \bigoplus a_i \quad \text{and} \quad \pi_i : \bigoplus a_i \to a_i$$

such that

$$\pi_i \circ \iota_i = id_{a_i} \forall i \quad \text{and} \quad \sum_i \iota_i \pi_i = \mathrm{id}_{\bigoplus a_i}$$

Recall that for semisimple categories, there is an isomorphism $\Phi : \bigoplus_{c,\mu} R_c \to R_a \otimes R_b$. Then we can build projection and injection maps in dependence of the tensor product by the following diagram:

$$\begin{tikzcd}
\bigoplus_c R_c \arrow[r, "\Phi"] & R_a \otimes R_b \arrow[r, "\Phi^{-1}"] \arrow[dr, "\exists \pi^{c,\mu}_{ab}"'] & \bigoplus_c R_c \arrow[d, "\pi_c"] \\
R_c \arrow[u, "\iota_c"] \arrow[ur, "\exists \iota^{ab}_{c,\mu}"'] & & R_c
\end{tikzcd} \tag{14}$$

Furthermore, these admit similar constraints as the direct sum:

$$\pi^{c,\mu}_{ab} \iota^{ab}_{d,\nu} = \pi_{c,\mu} \Phi^{-1} \Phi \iota_{d,\nu} = \delta_{cd} \delta_{\mu\nu} \, \mathrm{id}_{R_c} \tag{15}$$

and

$$\sum_{c,\mu} \iota^{ab}_{c,\mu} \pi^{c,\mu}_{ab} = \Phi \left(\sum_{c,\mu} \iota_{c,\mu} \pi_{c,\mu} \right) \Phi^{-1} = \Phi \, \mathrm{id}_{\bigoplus R_c} \, \Phi^{-1} = \mathrm{id}_{R_a \otimes R_b} \tag{16}$$

Definition 6.7 (Enriched Category). Let $(\mathcal{V}, \otimes, I, \alpha, \lambda, \rho)$ be a monoidal category. A category C *enriched over* $\mathcal{V}$ (or a $\mathcal{V}$*-category*) consists of

1. a set of objects $\mathrm{ob}(C)$,
2. for every $A, B \in \mathrm{ob}(C)$, a *hom-object* $C(A, B) \in \mathcal{V}$ (rather than *Set*),
3. for every triple $A, B, C \in \mathrm{ob}(C)$, a *composition morphism* in $\mathcal{V}$

$$\circ_{A,B,C} : C(B, C) \otimes C(A, B) \longrightarrow C(A, C),$$

4. for every $A \in \mathrm{ob}(C)$, a *unit morphism* in $\mathcal{V}$

$$j_A : I \longrightarrow C(A, A),$$

6.2 Categorical Quantum Mechanics

By a result of Abramsky [Abr12], building on the theory of coecke et al., we can state the no-cloning theorem in categorical form: This asserts that there is no diagonal $\delta : V \to V \otimes V$ such that the following diagram commutes:

$$\begin{tikzcd} V \arrow[r, "f"] \arrow[d, "\delta"'] & W \arrow[d, "\delta"] \\ V \otimes V \arrow[r, "f \otimes f"] & W \otimes W \end{tikzcd}$$

6.3 Decompositions for Categorical Networks

This chapter generalizes the decomposition in [SV12, Equation 27] to both Fusion Categories and the Representation categories of compact groups. Note that in [DH25], such a generalization is given for representations of finite groups, which does not include $SU(2)$. However I do not yet know if we should do so, since I not yet sure how to argue the F moves here. If we instead take the deformation $U_q(sl_2)$ we get finiteness (and fusion cats?), unfortunately $U_q(sl_2)$ is not a group.

6.4 Decomposition of semisimple Categories

Lemma 6.8 (Isotypic Decomposition)**.** *Let C be a k-enriched semisimple category. Then for every $V \in C$, writing $D_a \coloneqq \mathrm{Hom}_C(R_a, V)$ for its* degeneracy spaces, *only finitely many D_a are nonzero, each is finite-dimensional, and there is a canonical isomorphism*

$$\bigoplus_{a \in L} D_a \otimes R_a \xrightarrow{\sim} V$$

with $D_a \otimes R_a$ as in Lemma **??**.

Proof. Observe that $V = \bigoplus_i R_a^{\oplus m_a}$ and

$$Hom_c(R_a, V) = \bigoplus_b Hom(R_a, R_b^{\oplus m_b})$$

□

Lemma 6.9. *For a vector space W, there is an isomorphism*

$$\bigoplus^m W \simeq \mathbb{C}^m \otimes W.$$

Proof. Let $e_{i\, i \in [m]}$ be the standard basis on $\mathbb{C}^m$. Let

$$\Phi : \mathbb{C}^m \otimes W \to \oplus^m W$$

be defined by

$$\phi(e_i \otimes w) = (0, \ldots, w, \ldots 0)$$

and linear, thus

$$\Phi\left(\sum_i c_i e_i \otimes w_i\right) = (c_1 w_1, \ldots, c_m w_m).$$

is clearly bijective, and we have our desired isomorphism. □

Lemma 6.10. *Let D be a finite dimensional k-vector space and let V be an object of a k-linear category C. Then*

$$D \otimes V \cong V^{\oplus \dim D}$$

Proof. Let $n = \dim D$. Take a basis $\{e_i\}$ for D. This yields an isomorphism

$$D \cong \bigoplus_{i=1}^{n} k \cdot e_i \cong k^n$$

Since in a k-linear category the tensor product is additive (should we prove this?), we can distribute

$$D \otimes W \cong \bigoplus_{1}^{n} (k \cdot e_i \otimes W) \cong \bigoplus W = W^{\oplus n}$$

□

Lemma 6.11 (Isotopic Decomposition). *Let C be a semisimple k-linear category with simple objects $\{R_a\}$. Then every object $V \in C$ admits a canonical decomposition*

$$V \cong \bigoplus_{a} D_a \otimes R_a. \tag{17}$$

where we refer to $D_a \coloneqq \mathrm{Hom}_C(R_a, V)$ as the degeneracy *of the space.*

Proof. Since C is semisimple, there exist integers $m_a \geq 0$ such that

$$V \cong \bigoplus_{a \in L} \bigoplus_{\_=1}^{D_a^V} a \cong \bigoplus_{a \in L} R_a^{\oplus m_a}.$$

Applying the additive functor $\mathrm{Hom}_C(R_b, -)$ yields

$$\mathrm{Hom}_C(R_b, V) \cong \bigoplus_{a \in L} \mathrm{Hom}_C(R_b, R_a^{\oplus m_a}) \cong \bigoplus_{a \in L} \mathrm{Hom}_C(R_b, R_a)^{\oplus m_a}.$$

Then, by Schurs lemma

$$\bigoplus_{a \in L} Hom_C(R_b, R_a)^{\oplus m_a} = k^{\oplus m_b}$$

□

So we actualy need this: Maybe this is simpler if we always let the degeneracy be C?

Lemma 6.12. *Let C be a k-linear category and let D, E be finite-dimensional k-vector spaces. For any objects $R, S \in C$, there is a natural isomorphism*

$$\mathrm{Hom}_C(D \otimes R, E \otimes S) \cong \mathrm{Hom}_k(D, E) \otimes \mathrm{Hom}_C(R, S).$$

Proof. Let $m = \dim_k D$ and $n = \dim_k E$. Proof by isomorphism chains:

$$\begin{aligned}
\mathrm{Hom}_C(D \otimes R, E \otimes S) &\cong \mathrm{Hom}_C(R^{\oplus m}, S^{\oplus n}) && \text{by lemma 6.9} \\
&\cong \bigoplus_{i=1}^{m} \bigoplus_{j=1}^{n} \mathrm{Hom}_C(R, S) && \text{by additivity of } \mathrm{Hom}_C \\
&\cong k^{mn} \otimes \mathrm{Hom}_C(R, S) && \text{by lemma 6.9} \\
&\cong \mathrm{Hom}_k(D, E) \otimes \mathrm{Hom}_C(R, S) && \text{since } \mathrm{Hom}_k(D, E) \cong k^{mn}
\end{aligned}$$

(The last as s) □

Definition 6.13 (Copower). For a $\mathrm{Vect_k}$-enriched (and tensored?) category C, and $E \in \mathrm{Vect_k}, X \in C$ the power X^E is the object of C representing the natural isomorphism

$$\mathrm{Hom}_C(Y, X^E) \cong \mathrm{Hom}_k(E, \mathrm{Hom}_C(Y, X))$$

Theorem 6.14 (Decomposition of Semisimple Categories). *Let C be a semisimple, monoidal, k-linear category whose tensor product is k-linear in each variable, with simple objects $\{R_a\}_{a\in L}$ and let V_i, W_i be a collection of objects in C. Every morphism*

$$T \in \mathrm{Hom}_C\left(\bigotimes_{i=1}^{n} V_i, \bigotimes_{i=1}^{m} W_i\right)$$

admit a decomposition as

$$T \in \bigoplus_{\hat{a},\hat{b}} P^{\hat{a},\hat{b}} \otimes Hom_C(R_{\hat{a}}, R_{\hat{b}}),$$

where $\hat{a} = (a_1, \ldots, a_n)$ and P is a k-vector space of dimension ???

Proof.

$$\begin{aligned}
&\mathrm{Hom}_C\left(\bigotimes_i V_i, \bigotimes_j W_j\right) \\
&\cong \mathrm{Hom}_C\left(\bigotimes_i \Big(\bigoplus_{a_i} D_{i,a_i} \otimes R_{a_i}\Big), \bigotimes_j \Big(\bigoplus_{b_j} E_{j,b_j} \otimes R_{b_j}\Big)\right) && \text{(Lemma 6.11)} \\
&\cong \mathrm{Hom}_C\left(\bigoplus_{\hat{a}} D_{\hat{a}} \otimes R_{\hat{a}}, \bigoplus_{\hat{b}} E_{\hat{b}} \otimes R_{\hat{b}}\right) && \text{(linearity of } \otimes) \\
&\cong \bigoplus_{\hat{a},\hat{b}} \mathrm{Hom}_C\big(D_{\hat{a}} \otimes R_{\hat{a}}, E_{\hat{b}} \otimes R_{\hat{b}}\big) && \text{(additivity of } \mathrm{Hom}_C) \\
&\cong \bigoplus_{\hat{a},\hat{b}} \mathrm{Hom}\big(D_{\hat{a}}, E_{\hat{b}}\big) \otimes \mathrm{Hom}_C\big(R_{\hat{a}}, R_{\hat{b}}\big) && \text{(Lemma 6.12)}
\end{aligned}$$

□

Corollary 6.15 (Operators are block matrices [KZ22]). *thm header. An operator*

$$Hom(V, W) \cong Hom_C\Big(\bigoplus_a R_a^{\oplus d_a}, \bigoplus_b R_b^{\oplus e_b}\Big) \cong \bigoplus_{a,b} Hom(D_a, E_b) \otimes Hom_C(R_a, R_b)$$

and by Schurs lemma this is just

$$\bigoplus_{a\in Irr(V)\cap Irr(W)} Hom(D_a, E_a) \cong \bigoplus_{a\in Irr(V)\cap Irr(W)} Mat_{d_a\times e_a}(k)$$

Corollary 6.16. *Let $Q_{\mu,\eta}$ be a basis of $Hom_C(\bigotimes_{\hat{a}} R_{\hat{a}}, \bigotimes_{\hat{b}} R_{\hat{b}})$ where μ denotes the multiplicities and η the internal fusions. In a* multiplicity-free *Category we will omit μ as a label.*

Example 6.17. However, now assign $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ to each leg in the example above. Each leg then becomes $V_0 \oplus V_1 \cong \mathbb{C} \otimes V_0 \oplus \mathbb{C} \otimes V_1$. Then we can start with the decomposition.

$$\begin{aligned}
&Hom_C\Big(\bigoplus_{a,b,c\in(0,1)} (C\otimes C\otimes C) \otimes V_a \otimes V_b \otimes V_c, \bigoplus_{b\in(0,1)} C \otimes V_b\Big) \\
&= \bigoplus_{a,b,c,d\in(0,1)} Hom_C((C\otimes C\otimes C) \otimes (V_a \otimes V_b \otimes V_c), C \otimes V_d) \\
&= \bigoplus_{a,b,c,d} Hom(C^3, C) \otimes Hom_{\mathrm{Rep}(SU(2))}(V_a \otimes V_b \otimes V_c, V_d)
\end{aligned}$$

Example 6.18. Easier. However, now assign $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ to each leg in the example above. Each leg then becomes $V_0 \oplus V_1 \cong \mathbb{C} \otimes V_0 \oplus \mathbb{C} \otimes V_1$. Then we can start with the decomposition.

$$
\begin{aligned}
&Hom_C\left(\bigoplus_{a,b\in(0,1)} (C \otimes C) \otimes V_a \otimes V_b \bigoplus_{b\in(0,1)} C \otimes V_b\right) \\
&= \bigoplus_{a,b,c\in(0,1)} Hom_C((C \otimes C) \otimes (V_a \otimes V_b), C \otimes V_c) \\
&= \bigoplus_{a,b,c} Hom(C^2, C) \otimes Hom_{\mathrm{Rep}(SU(2))}(V_a \otimes V_b, V_c)
\end{aligned}
$$

Example 6.19. Lets look at a tensor with in legs $(\tau \otimes \tau)$ and τ and out legs τ.

$$Hom_C(((k \otimes 1) \oplus (k \otimes \tau)) \otimes (k \otimes \tau), k \otimes \tau) = Hom_C()$$

$$
\begin{aligned}
&\mathrm{Hom}_C\big(((k \otimes 1) \oplus (k \otimes \tau)) \otimes (k \otimes \tau),\ k \otimes \tau\big) \\
&\cong \mathrm{Hom}_C\big((k \otimes 1 \otimes k \otimes \tau) \oplus (k \otimes \tau \otimes k \otimes \tau),\ k \otimes \tau\big) \\
&\cong \mathrm{Hom}_C(k \otimes 1 \otimes k \otimes \tau,\ k \otimes \tau) \oplus \mathrm{Hom}_C(k \otimes \tau \otimes (k \otimes \tau),\ k \otimes \tau) \\
&\cong \mathrm{Hom}(k \otimes k, k) \otimes \mathrm{Hom}_C(1 \otimes \tau, \tau) \oplus \mathrm{Hom}(k \otimes k, k) \otimes \mathrm{Hom}_C(\tau \otimes \tau, \tau).
\end{aligned}
$$

Example 6.20 (Anyonic matrix operator)**.** Finally, consider a $4 \to 4$ anyon operator with one leg on each side, representing the space $\mathrm{Hom}((\tau \otimes \tau \otimes \tau \otimes \tau), (\tau \otimes \tau \otimes \tau \otimes \tau))$. THe full decomposition is $2 + 3\tau$. Hence we obtain

$$\mathrm{Hom}_C\big((k^2 \otimes 1) \oplus (k^3 \otimes \tau), (k^2 \otimes 1) \oplus (k^3 \otimes \tau)\big)\,.$$

$$= \bigoplus_{a,b\in(1,\tau)} \mathrm{Hom}_C(k^{m_a} \otimes a,\ k^{m_b} \otimes b) = \bigoplus_{a,b\in(1,\tau)} \mathrm{Hom}(k^{m_a}, k^{m_b}) \otimes Hom_C(a, b)$$

where

$$m_1 = 2, \qquad m_\tau = 3.$$

For this case, Schurs lemma applies. So we have

$$= Hom(k^2, k^2) \oplus Hom(k^3, k^3)$$

which is just a block matrix with indexing $(1_1, 1_2, \tau_1, \tau_2, \tau_3)$.

Example 6.21 ($U(1)$ Symmetry)**.** Technically we would want to show here that this is simpler in general. TOOD. Make a theorem? Or a corollary?

6.5 Tensor and Fusion Categories

Let C and D be linear abelian categories. Then the *Deligne Tensor Product* $C \boxtimes D$ is a linear abelian category with a bifunctor

$$\boxtimes : C \times D \to C \boxtimes D$$

with morphisms given by

$$Hom_{C\boxtimes D}(X \boxtimes Y, W \boxtimes V) = Hom_C(X, W) \otimes Hom_D(Y, V)$$

Definition 6.22 (The Deligne Tensor Product of Categories)**.** Let k be a field, and let C and $\mathcal{D}$ be two k-linear, finite abelian categories. The **Deligne tensor product**, denoted by $C \boxtimes \mathcal{D}$, is a k-linear finite abelian category equipped with a k-bilinear functor

$$\boxtimes : C \times \mathcal{D} \longrightarrow C \boxtimes \mathcal{D}, \quad (X, Y) \mapsto X \boxtimes Y$$

TODOTOD!!

Definition 6.23 (The Category $\mathrm{Rep}_k(G)$)**.** Let G be a group and k a field. The *category of linear representations of G over k* consists of the following data:

- **Objects:** The objects $\mathrm{Rep}_k(G)$ are pairs (V, ρ_V), where V is a k-vector space and $\rho_V : G \to \mathrm{GL}(V)$.
- **Morphisms:** For any two objects (V, ρ_V) and (W, ρ_W) in $\mathrm{Rep}_k(G)$, a morphism from V to W is a k-linear transformation $f : V \to W$ that is G-invariant (also called an *intertwiner*). That is, for every element $g \in G$, the following diagram commutes:

$$\begin{tikzcd} V \arrow[r, "\rho_V(g)"] \arrow[d, "f"'] & V \arrow[d, "f"] \\ W \arrow[r, "\rho_W(g)"'] & W \end{tikzcd}$$

  In algebraic terms, f must satisfy the condition:

$$f \circ \rho_V(g) = \rho_W(g) \circ f \quad \forall g \in G$$

  The set of all such morphisms is denoted by $\mathrm{Hom}_G(V, W)$.

We can further endow $Rep_k(G)$ with a monoidal structure: For any two objects $(V, \rho_V), (W, \rho_W)$, their tensor product is in $\mathrm{Rep}_k(G)$ by the diagonal action

$$\rho_{(V \otimes W)}(g)(v \otimes w) = \rho_V(g)v \otimes \rho_W(g)w$$

Theorem 6.24 (Fusion Representation Category of finite groups)**.** *Let G be a finite group. Then* $\mathrm{Rep}(G)$ *is a fusion category.*

Theorem 6.25. *Let G_1 and G_2 be finite groups and let* $\mathrm{Rep}(G_1), \mathrm{Rep}(G_2)$ *be their respective symmetry categories. Then there is an equivalence of categories*

$$\mathrm{Rep}(G_1 \times G_2) \simeq \mathrm{Rep}(G_1) \boxtimes \mathrm{Rep}(G_2)$$

Lemma 6.26 (Schur's lemma [Sch05])**.** *If S and T are simple objects in a semisimple category over an algebraically closed field (this is always C in our case), then it holds that*

$$\mathrm{Hom}(S, T) = \begin{cases} k \cdot \mathrm{id}_S, & \text{if } S \cong T; \\ 0, & \text{if } S \not\cong T. \end{cases}$$

Proof [IR19, Theorem 7.5].

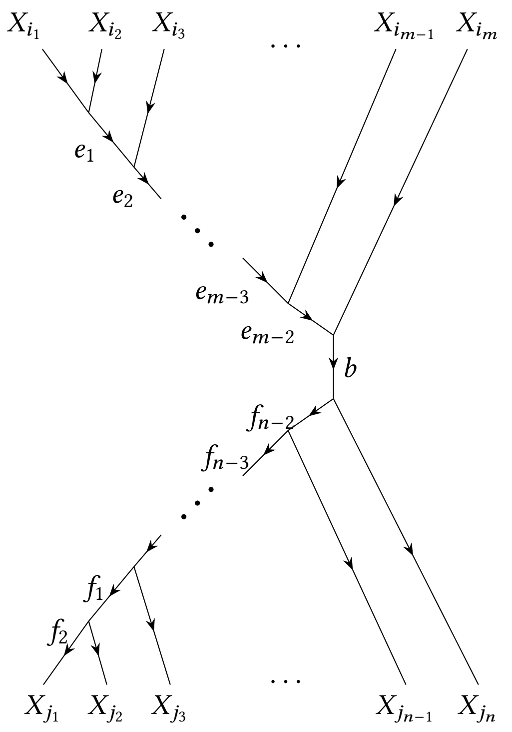
Lets give a coordinatized approach to fusion categories:

Definition 6.27 (Fusion Category in Vector Form)**.** A *fusion category* is a finite set of labels (sectors?) L and vector spaces V_c^{ab} for all $a, b, c \in L$, as well as the isomorphisms

$$F_d^{abc} : \bigoplus_e V_e^{ab} \otimes V_d^{ec} \to \bigoplus_e V_d^{ae} \otimes V_e^{bc}$$

such that the pentagon holds.

$$\mathrm{Hom}\left(\bigotimes_{k=1}^{m} X_{i_k}, \bigotimes_{l=1}^{n} X_{j_l}\right) = \mathrm{span}_{\mathbb{C}} \left\{ \quad \right\}_{\substack{b,\, e_1, \ldots, e_{m-2},\, f_1, \ldots, f_{n-2} \\ \mu_1, \ldots, \mu_{m-2},\, \nu_1, \ldots, \nu_{n-2}}} \tag{18}$$



$$\begin{tikzcd}
 & & \bigoplus\limits_{e,l} V_e^{ab} \otimes V_f^{cd} \otimes V_g^{el} \arrow[drr, "F_g^{abl} \otimes id_{V_l^{cd}}"] & & \\
\bigoplus\limits_{e,f} V_e^{ab} \otimes V_f^{ec} \otimes V_g^{fd} \arrow[urr, "id_{V_e^{ab}} \otimes F_g^{ecd}"] \arrow[dr, "F_f^{abc} \otimes id_{V_g^{fd}}"'] & & & & \bigoplus\limits_{k,l} V_g^{ak} \otimes V_k^{bl} \otimes V_l^{cd} \\
 & \bigoplus\limits_{f,h} V_h^{af} \otimes V_f^{bc} \otimes V_g^{hd} \arrow[rr, "F_g^{afd} \otimes id_{V_f^{bc}}"'] & & \bigoplus\limits_{k,h} V_g^{ak} \otimes V_h^{bc} \otimes V_k^{hd} \arrow[ur, "F_k^{bcd} \otimes id_{V_g^{ak}}"'] &
\end{tikzcd} \tag{19}$$

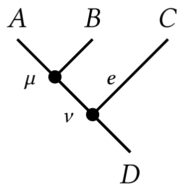
$$\begin{tikzcd}
 & \bigoplus\limits_{f,h} V_h^{af} \otimes V_f^{bc} \otimes V_g^{hd} \arrow[dr] & \\
\bigoplus\limits_{e,f} V_e^{ab} \otimes V_f^{ec} \otimes V_g^{fd} \arrow[ur] \arrow[d, "F_f^{abc} \otimes id_{V_g^{fd}}"'] & & \bigoplus\limits_{f,h} V_h^{af} \otimes V_f^{bc} \otimes V_g^{hd} \\
\bigoplus\limits_{f,h} V_h^{af} \otimes V_f^{bc} \otimes V_g^{hd} \arrow[rr, "F_f^{abc} \otimes id_{V_g^{fd}}"'] & & \bigoplus\limits_{k,h} V_g^{ak} \otimes V_h^{bc} \otimes V_k^{hd} \arrow[u]
\end{tikzcd}$$

$$\begin{tikzcd}
V \times W \arrow[r, "\phi"] \arrow[dr, "f"'] & V \otimes W \arrow[d, dashed, "\exists!\, \tilde{f}"] \\
 & U
\end{tikzcd} \tag{20}$$

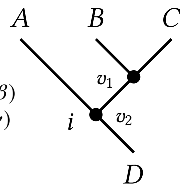
I think we may be able to build a nice summary with this diagram:

$$
\begin{tikzcd}[column sep=large]
(A\otimes B)\otimes C \arrow[r, "\alpha_{ABC}", "\sim"'] \arrow[d, "Hom(-{,}D)"] & A\otimes(B\otimes C) \arrow[d, "Hom(-{,}D)"] \\
Hom((A\otimes B)\otimes C, D) \arrow[r, "\alpha_{A,B,C}", "\text{associator induces isomorphims under yoneda}"'] \arrow[d, phantom, "\cong" , "\text{semisimplicity}"'] & Hom(A\otimes(B\otimes C), D) \arrow[d, phantom, "\cong", "\text{semisimplicity}"'] \\
\bigoplus_{c\in Irr(CAT)} Hom(A\otimes B, c)\otimes Hom(c\otimes C, D) \arrow[r, phantom, "\cong"] \arrow[d, phantom, "\parallel"] & \bigoplus_{d\in Irr(\mathrm{Cat})} Hom(A\otimes d, D)\otimes Hom(B\otimes C, d) \arrow[d, phantom, "\parallel"] \\
\bigoplus_{c} V_c^{AB}\otimes V_D^{cC} \arrow[r, phantom, "\cong"] \arrow[d, phantom, "\Cap"] & \bigoplus_{c} V_D^{Ac}\otimes V_c^{BC} \arrow[d, phantom, "\Cap"] \\
|(A,B);c;\mu\rangle \otimes |c,C;D;\nu\rangle \arrow[r, "F_D^{ABC}", "\text{6j-symbols}"'] & \sum_{f,\alpha,\beta} [F_D^{ABC}]^{(f\alpha\beta)}_{(c\mu\nu)} |A,f;D;\beta\rangle \otimes |B,C;f;\alpha\rangle
\end{tikzcd}
$$

$\mu\in 1,\ldots,N_c^{AB}$ $\quad$ $\nu\in 1,\ldots,N^{cC,D}$



$$= \sum_{f,\alpha,\beta} [F_D^{ABC}]^{(f\alpha\beta)}_{(e\mu\nu)}$$



A Appendix

A.1 Proofs for Section 3

This is the proof for statement Equation (5) on page 8.

Proof. Recall that when two operators commute, we can *simultaneously diagonalize* them [HJ12, Theorem 1.3.12].

$$[H, S^z_{\text{tot}}] = \sum_i^N [\sigma_i^x\sigma_{i+1}^x + \sigma_i^y\sigma_{i+1}^y + \Delta\sigma_i^z\sigma_i^z, \sum_{j=1}^N \sigma_j^z]$$

and since operators on different sites commute, only $j = i$ and $j = i+1$ contribute. So we ought to write

$$\sum_i^N [\sigma_i^x\sigma_{i+1}^x + \sigma_i^y\sigma_{i+1}^y + \Delta\sigma_i^z\sigma_i^z, \sigma_i^z + \sigma_{i+1}^z]$$

and now using the identity $[AB, C] = A[B, C] + [A, C]B$:

$$[\sigma_i^x\sigma_{i+1}^x, \sigma_i^z + \sigma_{i+1}^z] = -2i(\sigma_i^y\sigma_{i+1}^x + \sigma_i^x\sigma_{i+1}^y)$$

$$[\sigma_i^y\sigma_{i+1}^y, \sigma_i^z + \sigma_{i+1}^z] = 2i(\sigma_i^x\sigma_{i+1}^y + \sigma_i^y\sigma_{i+1}^x)$$

which cancel out. Furtermore, the z spins commute as well. Thus the whole sum is 0 and we are done. □

A.2 Supplementary categorical constructions

Note on universal properties: When dealing with categories, we are fundamentally dealing with abstractions that behave alike to a known construction. These behaviours are made precise by the use of a *universal property*. We illustrate this with the definition of a (co)product:

Definition A.1. Let C be a category and $x, y \in ob(C)$. Their *product* is an object $x \prod y$ in C with morphisms $\pi_x : x \prod y \to x$ and $\pi_y : x \prod y \to y$. Their *coproduct* is defined as the object $x \coprod y$ together with the maps $\iota_x : x \to x \coprod y$ and $\iota_y : y \to x \coprod y$. Both of these definitions are universal with their properties.

Definition A.2 ([Mos25]). A *fusion category* can be restated in its vector space form as a finite set of labels (sectors?) L and vector spaces V_c^{ab} for all $a, b, c \in L$, as well as the isomorphisms

$$F_d^{abc} : \bigoplus_e V_e^{ab} \otimes V_d^{ec} \to \bigoplus_e V_d^{ae} \otimes V_e^{bc}$$

such that the pentagon holds:

$$\begin{tikzcd}
 & & \bigoplus_{e,l} V_e^{ab} \otimes V_f^{cd} \otimes V_g^{el} \arrow[drr, "F_g^{abl} \otimes id_{V_l^{cd}}"] & & \\
\bigoplus_{e,f} V_e^{ab} \otimes V_f^{ec} \otimes V_g^{fd} \arrow[urr, "id_{V_e^{ab}} \otimes F_g^{ecd}"] \arrow[dr, "F_f^{abc} \otimes id_{V_g^{fd}}"'] & & & & \bigoplus_{k,l} V_g^{ak} \otimes V_k^{bl} \otimes V_l^{cd} \\
 & \bigoplus_{f,h} V_h^{af} \otimes V_f^{bc} \otimes V_g^{hd} \arrow[rr, "F_g^{afd} \otimes id_{V_f^{bc}}"'] & & \bigoplus_{k,h} V_g^{ak} \otimes V_h^{bc} \otimes V_k^{hd} \arrow[ur, "F_k^{bcd} \otimes id_{V_g^{ak}}"'] &
\end{tikzcd} \tag{21}$$

A.3 Overview of existing implementations

Implementations that are available:

Name	**Symmetry**	**Citation**(Software, Publication)	Language
ITensor	Nonabelian (Fermion?)	[FWS22]	Julia
Cyten	Categorical	[UH] [Unf24]	python
TeNPy	Abelian	[Hau+24]	python
QSpace	Nonabelian	[Wei24] [Wei12a]	?
ChemTensor	$SU(2)$	[Men26]	C
UniTensor	?	[**<empty citation>**]	? (Python, C++)
YasTn	Abelian	[Ram+25]	Python
TensorKit.jl	Categorical	[DH25]	Julia
Cytnx	Abelian only (i think)	[**<empty citation>**]	C++
quimb	Fermion?	[**<empty citation>**]	julia
MPSKit.jl	like TensorKit?	[**<empty citation>**]	julia
TeMFpy	Fermion?	[HS25]	python
abeliantensor	$U(1)$ and $\mathbb{Z}_n$	[Hau26]	python
symmray	Fermion	[Gao+25] [Gra26]	?
grassmanntn	Grassmann	[Yos26]	?
SymLibrary	? (OnSite)	[McM19] [MSB20]	MatLab?
Telum.jl	NonAbelian	[Lee26]	Julia

Table 1: Overview of tensor network libraries and symmetry support. (There are some closed source things here as well, also what is the software used by singh et al.?). Software is included if it has a website or documentation and an account of the symmetries it exploits. Applications or theory are listed in the Citation column.

Symbol	Meaning
$\mathbb{C}$	Complex numbers
$\mathbb{F}$	Base field
$\mathbb{N}$	Natural numbers
$\mathfrak{g}$	Lie algebra
$\mathfrak{sl}_2$	Special linear Lie algebra of degree 2
$U_q(\mathfrak{sl}_2)$	Quantum group associated with $\mathfrak{sl}_2$
$\mathcal{C}$	Category
$\mathbf{1}$	Monoidal unit object
V^*	Dual vector space/object
$\mathrm{Hom}(V, W)$	Space of morphisms from V to W
$\mathrm{Rep}(G)$	Representation category of group G
$\mathrm{Hom}_G(V, W)$	Space of intertwiners between representations of G
Vect_k	Category of k-vector spaces
$\mathrm{Hom}_k(-, -)$	Linear morphisms between k-vector spaces

Table 2: Notation conventions.

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